SYNTHETIC DATA — 교육용 가상 데이터이며 실제 제품 사양·운영 기준과 무관합니다.

Smart Power Meter 3P4W (METER-LP3500-3P4W) — Datasheet

1. Overview

Panel-mount multifunction power meter, class 0.5S, harmonics & logging. Product family: Power & energy meters. Lifecycle status: **ACTIVE**.

2. Technical specifications

2.1. Ratings

Item	Value
Accuracy class (active energy)	Class 0.5S (IEC 62053-22)
Voltage input	3P4W 57.7-400 V L-N
Current input	CT secondary 5 A or 1 A
Auxiliary supply	AC/DC 85-265 V

2.2. Performance and characteristics

Item	Value
Harmonic measurement	Up to 31st order
Data logging memory	8 MB (approx. 1 year at 15-min interval)

2.3. Endurance and communication

Item	Value
Communication	Modbus RTU (RS-485), Modbus TCP option

3. Installation data

3.1. Terminals

- Dimensions (W x H x D): 96 x 96 x 70 mm
- Weight: 0.45 kg
- Degree of protection: IP54 (front)
- Operating temperature: -10 to +55 °C

4. Standards and certifications

Applicable standards:

- IEC 62053-22
- IEC 61557-12

Certifications held:

- CE
- KC

Certificates and declarations of conformity are available from the Certification & Compliance team on request. Certifications not listed above are not held for this product; project-specific evaluation can be requested.

5. Ordering information

Item	Value
Ordering code	METER-LP3500-3P4W
Standard lead time	3 weeks ex-works
Minimum order quantity	5
Warranty	24 months (see POL-WARRANTY)
Firmware	latest 1.80 (see RN-METER-LP3500-3P4W)

6. Related documents

- IG-METER-LP3500-3P4W — Installation and Commissioning Guide
- TG-METER-LP3500-3P4W — Troubleshooting and Error Code Guide
- RN-METER-LP3500-3P4W — Firmware Release Notes
- FAQ-METER — Power & energy meters FAQ

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Smart Power Meter 3P4W (METER-LP3500-3P4W) — Installation and Commissioning Guide

1. Safety

Installation, wiring and commissioning must be performed by qualified personnel. Disconnect and lock out all power sources before work. Verify absence of voltage.

2. Receiving and storage

Inspect the packaging on arrival and report damage within 7 days (see POL-RETURN). Store indoors in a dry place between -25 and +55 °C, relative humidity below 85% without condensation.

3. Mounting and clearances

- Required clearances: Panel cut-out 92 x 92 mm (panel type) or 35 mm DIN rail
- Operating temperature: -10 to +55 °C
- Degree of protection: IP54 (front)
- Above 1,000 m altitude, apply derating per the datasheet.

4. Wiring and tightening torques

- Terminal tightening torque: see dimension drawing
- Use copper conductors rated for at least 75 °C.
- Separate power and signal cables by at least 200 mm.

5. Commissioning procedure

1. Check CT polarity and phase sequence
2. Set CT and VT ratios
3. Set communication address and baud rate
4. Verify readings against a reference meter

6. Pre-energisation checklist

☐ Product rating matches the drawing
☐ All terminals tightened to specified torque
☐ Protective earth connected
☐ Insulation resistance measured and recorded
☐ Settings / parameters recorded

7. Common commissioning issues

Symptom	Likely cause	See

Negative active power reading	CT polarity reversed	TG-METER-LP3500-3P4W §3.1
No Modbus response	Address / baud / parity mismatch	TG-METER-LP3500-3P4W §3.2
No response at 38400 even parity	Firmware defect at 38400 / even parity	TG-METER-LP3500-3P4W §3.3
Energy register reset after power loss	Firmware defect during flash write	TG-METER-LP3500-3P4W §3.4

On-site commissioning support can be requested through the Global CS Center. For power & energy meters the request is handled by the technical support team.

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Smart Power Meter 3P4W (METER-LP3500-3P4W) — Firmware Release Notes

1. Summary

Latest firmware: **1.80**. Updating to the latest version is recommended for all units.

2. Version history

2.1. Version 1.80 (2025-12-01)

- Added: Modbus TCP gateway mode
- Fixed: R.01 no response when baud 38400 and parity even

2.2. Version 1.72 (2025-04-07)

- Fixed: energy register reset after power loss during flash write

2.3. Version 1.60 (2024-08-05)

- Baseline release

3. Update procedure

1. Back up parameters / project before updating.
2. Connect the engineering tool (v4.6 or later).
3. Do not switch off power during the update.
4. After the update, verify the version and restore parameters if required.
5. If the update fails, do not power-cycle the unit repeatedly; contact support for the recovery procedure.

4. Compatibility

Parameter files from older versions can be loaded; files from newer versions cannot be loaded into older firmware.